RESEARCH ARTICLE

Evolution of ESG-focused DLT research: An NLP analysis of the literature

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ABSTRACT

Distributed Ledger Technology (DLT) faces increasing environmental scrutiny, particularly concerning the energy consumption of the Proof of Work (PoW) consensus mechanism and broader Environmental, Social, and Governance (ESG) issues. However, existing systematic literature reviews of DLT rely on limited analyses of citations, abstracts, and keywords, failing to fully capture the field's complexity and ESG concerns. We address these challenges by analyzing the full text of 24,539 publications using Natural Language Processing (NLP) with our manually labeled Named Entity Recognition (NER) data set of 39,427 entities for DLT. This methodology identified 505 key publications at the DLT/ESG intersection, enabling comprehensive domain analysis. Our combined NLP and temporal graph analysis reveals critical trends in DLT evolution and ESG impacts, including cryptography and peer-to-peer networks research's foundational influence, Bitcoin's persistent impact on research and environmental concerns (a "Lindy effect"), Ethereum's catalytic role on Proof of Stake (PoS) and smart contract adoption, and the industry's progressive shift toward energy-efficient consensus mechanisms. Our contributions include the first DLT-specific NER data set addressing the scarcity of high-quality labeled NLP data in blockchain research, a methodology integrating NLP and temporal graph analysis for large-scale interdisciplinary literature reviews, and the first NLP-driven literature review focusing on DLT's ESG aspects.

1. INTRODUCTION

Emerging technologies are facing increased scrutiny regarding their energy consumption and broader ecological impacts, including the use of vital resources such as water, precious metals, and synthetic compounds (Platt, Sedlmeir et al., 2021; Simone, Petracci et al., 2022; Sun, Luo et al., 2024). This growing environmental awareness necessitates evaluating technological advancements through their ecological footprint, with Distributed Ledger Technology (DLT) being no exception. While DLT offers promising features like record immutability and decentralization, it also presents challenges, particularly the high energy consumption of specific consensus algorithms. For instance, Bitcoin's PoW (Nakamoto, 2008), designed to prevent Sybil attacks, in which malicious actors pretend to be multiple users, has significant energy requirements (Ibañez & Freier, 2023).

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The rapid evolution and interest in DLT have led to a growing number of publications and intensified scrutiny of its ecological footprint. Yet, persistent misconceptions about its energy consumption patterns have complicated objective assessment. Additionally, the complexity of DLT extends well beyond environmental considerations. DLT spans breakthroughs in security, cryptography, network design, and diverse applications, leading to a rapidly expanding body of research. Therefore, traditional systematic literature review methodologies and manual literature analysis techniques that primarily rely on citation metrics, abstract analyses, and database term searches demonstrate increasing insufficiency in capturing the multidimensional nature and full scope of the rapidly evolving DLT domain.

We address this complexity by turning to Natural Language Processing (NLP), a subfield of Artificial Intelligence (AI) and linguistics. NLP offers powerful tools to examine the growing body of DLT literature, from academic articles to industry whitepapers. Among its various applications, we focus on Named Entity Recognition (NER) to identify specific DLTs (e.g., Proof of Work (PoW) and Proof of Stake (PoS)) and their Environmental, Social, and Governance (ESG) implications (e.g., "energy consumption" and "computational power") within our data set. This approach allows us to understand the shifts in research across different DLTs and their environmental impacts.

Building on precedents from the biomedical field, where ontologies or hierarchical taxonomies are used to create NER data sets for NLP-driven literature mining and review (Alsheikh, Wollenhaupt et al., 2022; Chang, Chang et al., 2016; Huang, Lai et al., 2020; McEntire, Szalkowski et al., 2016; Nabi, Talib et al., 2022; Spasic, Ananiadou et al., 2005), we have successfully transferred these methods to the DLT domain. Unlike previous systematic literature reviews that rely on citation measures, abstract analyses, keywords, or database term searches, our approach examines the entire body of each publication. This allows us to capture all relevant entities under top-level categories representing the whole DLT field, removing the need for exhaustive database term searches to generate a data set representing the DLT field and enabling us to work at scale. By mapping each publication's tokens to the hierarchical DLT taxonomy from Tasca and Tessone (2019), which we extended to include ESG considerations and other categories, we detected thematic shifts in both academic research and industry publications (e.g., white papers). As a result, we compiled a more inclusive data set that reflects the DLT field's overall evolution while preserving the methodological rigor demonstrated in biomedical literature studies.

Our research makes the following contributions:

- Creating a manually labeled NER data set of 39,427 named entities for 12 top-level DLT taxonomy categories. To the best of our knowledge, this is the first NER data set explicitly designed for DLT. Our data set aims to address the NLP data scarcity issue in blockchain research. By making it openly available, we help to overcome common limitations like high labeling costs and complex access to high-quality labeled NLP data in the DLT field[1].
- Presenting a methodological framework for executing an NLP-driven systematic literature review at the intersection of domains, in this case, DLT and ESG research. Our methodology combines approaches from different publications and fields for domain-specific literature text mining and systematic literature reviews.
- Conducting the first NLP-driven systematic literature review for the DLT field that places a special emphasis on ESG aspects, deriving empirical insights into the evolution of the literature.

Our work, particularly our NER data set, can potentially support future NLP data-driven studies in the DLT field. Additionally, it represents a foundation for future research to improve

[1] The data set is available at: https://huggingface.co/datasets/ExponentialScience/ESG-DLT-NER.

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automated systematic literature review processes at scale, capable of capturing the intrinsic dependencies and evolution of concepts related to the intersection of fields.

2. RELATED WORK

- *Systematic reviews in DLT.* Previous literature reviews have extensively explored blockchain applications in various sectors (Casino, Dasaklis, & Patsakis, 2019; Zheng, Xie et al., 2018). These reviews, however, differ in scope and depth compared to our systematic review, particularly in terms of the number of analyzed articles. Previous studies have mainly focused on blockchain's role in decentralization and privacy and analyzed trends of centralization in decentralized systems such as Bitcoin and Ethereum (Sai, Buckley et al., 2021). Spychiger, Tasca, and Tessone (2021) deconstructed 107 blockchain technologies using a specific taxonomy, emphasizing consensus mechanisms and cryptographic primitives. Our work, in contrast, provides a broader perspective on the evolution of DLT, including its ESG implications, and uses a bigger corpus.
- *ESG implications.* Bilal, Malik et al. (2014), Jiang, Gu et al. (2022), Mengelkamp, Notheisen et al. (2018), Poberezhna (2018), Richardson and Xu (2020), Schulz and Feist (2021), and Wu, Fu et al. (2022) have explored blockchain's potential in energy management, environmental sustainability, carbon trading, and transparent reporting. Our study extends these approaches by examining the intersection of ESG and DLT through a literature analysis.
- *NLP applications.* Studies have demonstrated NLP's effectiveness in automated ESG scoring (Sokolov, Mostovoy et al., 2021) and opinion summarization (Dubey, Jain, & Kalaiselvi, 2023). Outside DLT, systematic reviews using NLP-driven methodologies, particularly in medical genomics (Alsheikh et al., 2022), have employed database term searches and abstract-based filtering, contrasting with our approach that uses a taxonomy, labeled NER data set, and language model to compile and filter a more inclusive corpus representing the DLT field and from which we derive insights.

2.1. Literature Reviews of DLT Application in Industry Sectors

- *Supply chain applications.* Blockchain applications in supply chain management have been extensively studied (Casella, Bigliardi et al., 2023; Ibañez, Bayer et al., 2025; Wannenwetsch, Ostermann et al., 2023), focusing on traceability, transparency, and process automation. These reviews highlight blockchain's role in enhancing supply chain visibility and reducing intermediary dependencies.
- *Internet of Things (IoT) integration.* Research examining blockchain-IoT integration (Alladi, Chamola et al., 2019; Dai, Zheng, & Zhang, 2019; Lo, Liu et al., 2019; Sallam, Qahtani, & Gaid, 2021) has emphasized security enhancement, data integrity, and decentralized device management. Studies have particularly focused on scalability challenges (Liu, Chen et al., 2025) and consensus mechanisms suitable for resource-constrained IoT devices.
- *Healthcare systems.* Literature reviews in healthcare applications (Ghosh & Das, 2020; Hussien, Yasin et al., 2019; Khatoon, 2020) have explored blockchain's potential for secure health records management, clinical trial data integrity, and pharmaceutical supply chain tracking.
- *Financial sector.* Blockchain-based shared ledgers have long been studied for their potential to impact accounting, auditing, and finance. In particular, the evolution of Decentralized Finance (DeFi) has generated systematic reviews (Chen & Bellavitis,

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2020; Cousaert, Xu, & Matsui, 2022; Gramlich, Guggenberger et al., 2023; Shah, Lathiya et al., 2023) examining blockchain's potential to reduce financial intermediation. This potential is exemplified through lending protocols (Xu & Vadgama, 2022), stablecoins (Ante, Fiedler et al., 2023), and Decentralized Exchanges (DEXs) (Xu, Paruch et al., 2023), which operate via smart contracts (Adams, Zinsmeister et al., 2021) or at the protocol level of a blockchain (Hernandez Cruz, Dahi et al., 2025; Peduzzi, James, & Xu, 2021). In particular, stablecoins (Ante et al., 2023) serve as the critical infrastructure bridging Traditional Finance (TradFi) and DeFi, though this interconnection can propagate market shocks between both (Hernandez Cruz, Xu et al., 2024).

3. METHODOLOGY

- *Data set construction and analysis approach*. Ontologies, specifically hierarchical taxonomies, are pivotal in developing NER data sets for text mining (Alsheikh et al., 2022; Chang et al., 2016; Huang et al., 2020; McEntire et al., 2016; Nabi et al., 2022; Spasic et al., 2005). For example, the GENIA corpus (Kim, Ohta et al., 2003), an NER data set from 2,000 biological abstracts, employs the GENIA ontology's hierarchical tree structure of 47 biological entities, including top-level categories such as biological source and substance to facilitate text mining in biomedical literature. Similarly, the Human Phenotype Ontology is used to create and expand NER data sets in biomedicine (Huang et al., 2020; Lobo, Lamurias, & Couto, 2017). Alsheikh et al. (2022), Chang et al. (2016), and McEntire et al. (2016) further demonstrate the use of ontology-based NER data sets for domain-specific literature text mining.

  Learning from these biomedical field precedents, our methodology (Figure 1) for NLP-based text mining and filtering (Figure 2) in the DLT field employs a hierarchical taxonomy (Tasca & Tessone, 2019) to manually annotate an NER data set of 80 full-text papers: 46 systematically reviewed publications of DLT's sustainability (Eigelshoven, Ullrich, & Bender, 2020) and 34 manually selected for their relevance to the ESG/DLT theme.
- *Two-tier analysis strategy*. Our analysis followed a two-tier approach to maximize the use of available data:

  1. *Broad analysis using metadata*: Keyword extraction (Rose, Engel et al., 2010) and topic modeling (Asmussen & Møller, 2019) provide a broad view of the main

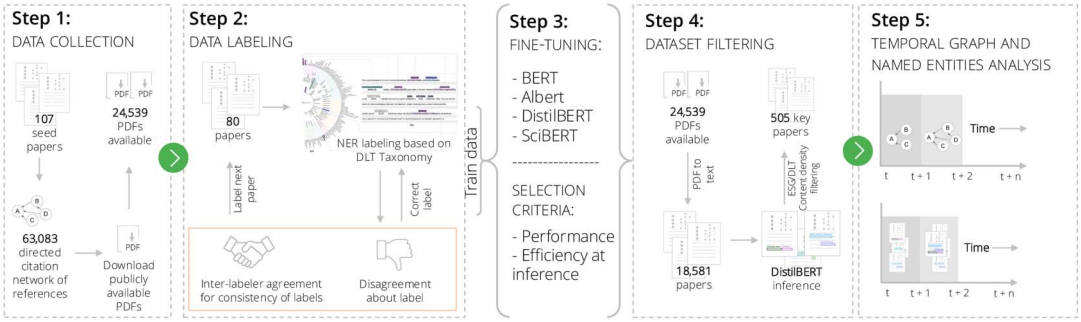


Figure 1. Methodology for the systematic literature review of ESG/DLT publications evolution using NER for content filtering, for which temporal graph and named entities (representing specific DLTs, e.g., PoW and PoS) analysis is carried.

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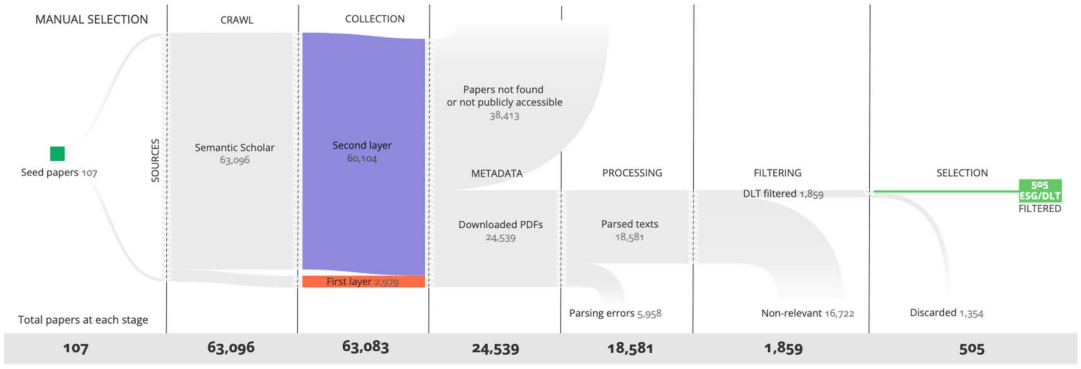


Figure 2. Processing pipeline for collecting and filtering papers in the review. The total number of papers present at each stage of processing is shown.

themes for a literature review. We analyzed the complete corpus of over 60,000 papers using their metadata (e.g., titles and keywords), regardless of full-text availability. This approach allows us to construct a comprehensive keywords graph that captures the broad evolution of the DLT field. While many papers (38,413) were behind paywalls or otherwise not easily accessible, their metadata still contributes to our understanding of the field's development.

2. *Deep analysis using full text*. We conducted a more detailed analysis of the subset of 24,539 papers where full text was available through open-access journals, conferences, or public repositories. We narrowed the number of publications for our analysis by fine-tuning a transformer-based language model for an NER task for ESG/DLT content density corpus filtering. This method allows us to categorize technologies in DLT, such as different `Consensus` mechanisms (e.g., Proof of Work, Proof of Stake) and their `ESG` implications, like "energy consumption" and "computational power," that would not usually emerge as single topics or keywords for a literature review. This deeper analysis reveals nuanced connections that may not arise from metadata alone.

- *Temporal graph and named entities analysis*. The final stage involves a temporal graph and named entities analysis to track the evolution of DLTs and their ESG implications over time. This combines insights from the broad metadata analysis and the detailed full-text examination, providing a comprehensive view of the field's development.

3.1. Data Collection

We constructed a directed citation network using a two-level expansion method: beginning with seed papers, collecting their references, and then gathering the references of those references. This two-level restriction helps maintain a thematic focus on ESG/DLT. The resulting network comprises 63,083 publications. Through Semantic Scholar's metadata for each publication (see Figure 1), we retrieved 24,539 full-text PDFs that are openly accessible through open-access journals, institutional repositories, preprint servers (e.g., arXiv), or publisher websites (see Figures 1 and 2).

The key benefit of using seed papers to build a citation network for a systematic literature review is the ease of expanding and updating the literature review by selecting seminal

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publications from the academic literature. The seed papers for our directed citation network were selected from two sources:

1. 89 papers from Eigelshoven et al. (2020), reviewing sustainability in popular DLT consensus algorithms.
2. 17 publications (2018–2022) with at least three citations each, chosen to update the corpus with more current research relevant to the ESG/DLT intersection (Alofi, Bokhari et al., 2022; Ante & Fiedler, 2021; Bada, Damianou et al., 2021; Cole & Cheng, 2018; Denisova, Mikhaylov, & Lopatin, 2019; Eshani, Rajdeep et al., 2021; Fernando & Saravannan, 2021; Ghosh & Das, 2020; Kohli, Chakravarty et al., 2022; Lucey, Vigne et al., 2022; Masood & Faridi, 2018; Platt et al., 2021; Powell, Hendon et al., 2021; Sapkota & Grobys, 2020; Schinckus, Nguyen, & Ling, 2020; Sedlmeir, Buhl et al., 2020, 2021).

3.2. NER Task for Literature Mining

We considered using Large Language Models (LLMs) for our NER task. An unsupervised approach like zero-shot NER with LLMs would have been more cost-effective in not requiring manually labeled data. However, at the time of this writing, domain-specific NER tasks often perform better with supervised learning models than with current LLMs (Hu, Chen et al., 2024; Li, Chan et al., 2023). This performance difference arises from fundamental architectural distinctions: Bidirectional Encoder Representations from Transformers (BERT)-based models generally outperform decoder-only LLMs, such as OpenAI's ChatGPT and Google Deepmind's Gemini, on token-level tasks like NER, due to their bidirectional encoder architecture (Devlin, Chang et al., 2019). Specifically, BERT processes text by simultaneously considering both left and right context (Devlin et al., 2019), enabling it to capture fine-grained contextual information around each token. This is a crucial capability for NER, where entity recognition often depends on the complete contextual surrounding of a word.

In contrast, decoder-only architectures, like Generative Pre-trained Transformer (GPT), process text unidirectionally (left to right) (Radford, Narasimhan et al., 2018; Radford, Wu et al., 2019), limiting their ability to capture the complete contextual information necessary for accurate entity recognition. While decoder-only architectures excel at open-ended question answering, they struggle to match BERT-based models' NER performance (Hu et al., 2024; Li et al., 2023), which currently represents the state of the art for NER tasks[2]. Therefore, we adopted a supervised learning approach, fine-tuning and benchmarking several BERT-base models, selecting the final model based on its performance and efficiency at inference. This approach provides more granularity for literature mining an entire publication body without the hallucination issues of LLMs (i.e., creation of irrelevant or inconsistent content). However, the trade-off is the requirement for the costly manual labeling of the NER data set.

3.2.1. Labeling

We manually annotated 80 publicly available publications using the brat tool (Stenetorp, Pyysalo et al., 2012) and argilla[3], following the extended taxonomy framework of Tasca

[2] https://paperswithcode.com/task/named-entity-recognition-ner.
[3] https://github.com/argilla-io/argilla.

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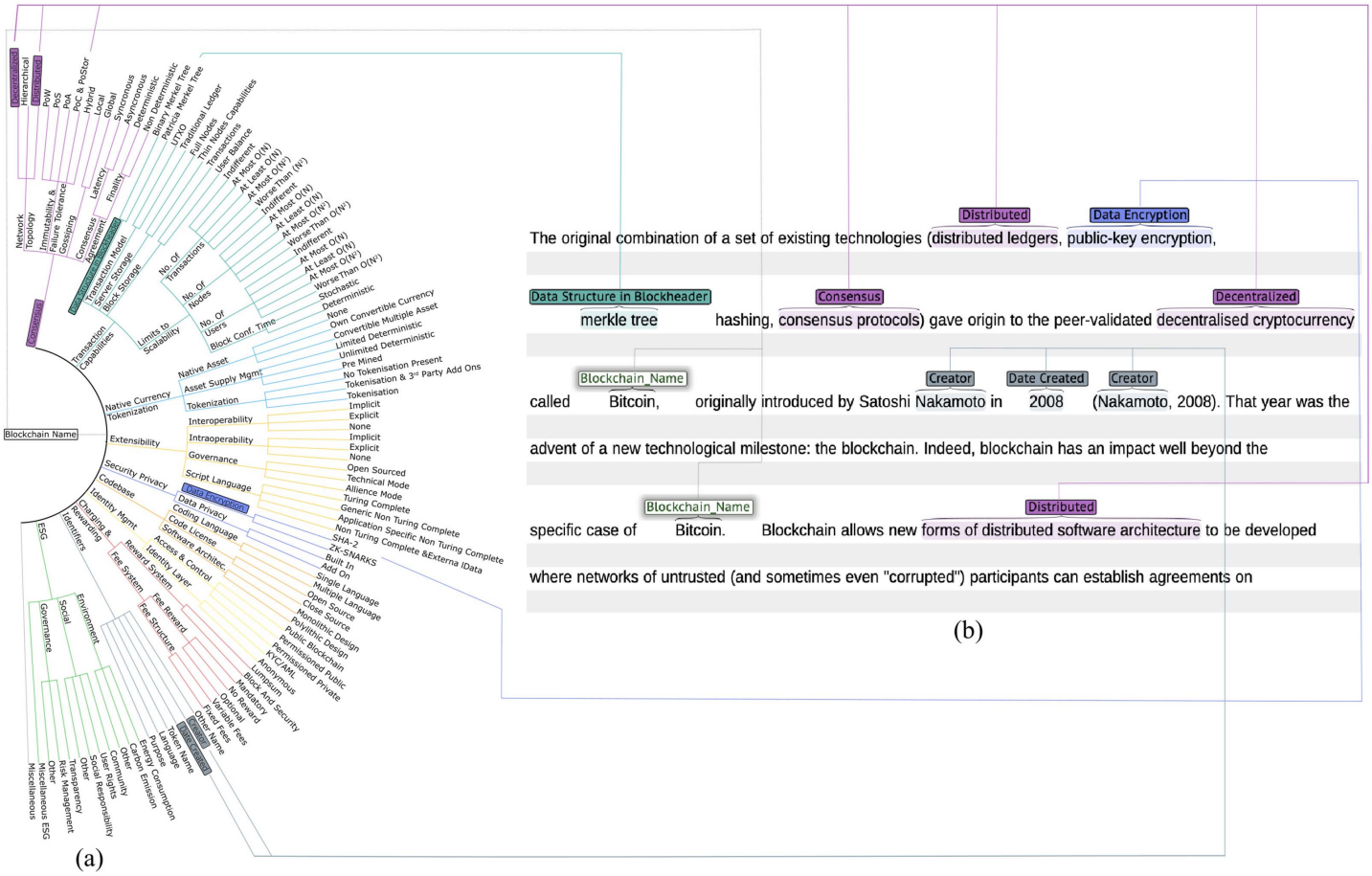


Figure 3. (a) The taxonomy of Tasca and Tessone (2019) extended with `Blockchain Name`, `ESG`, and `Miscellaneous` (see Section 3.2.1) for the purpose of this research. (b) Example of parsed text with the taxonomy label associated with a span of text labeled. The labels used in the paragraph are highlighted in the taxonomy tree.

and Tessone (2019) (Figure 3). This taxonomy provides a hierarchical structure of DLT technology components, with each principal component (e.g., `Consensus`) divided into subcomponents (e.g., `Gossiping`) and further into sub-sub-components if needed (e.g., `Local`). We introduced categories like `Blockchain Name` to identify specific blockchains. We expanded the initial definition of `Security Privacy` to label security threats (e.g., `Sybil attack` and `51% attack`). Also, we added the `Miscellaneous` category for ambiguous contexts (see Figure 3 and Table 1), following the example of the CoNLL-2003 data set for a similar category (Tjong, Sang, & De Meulder, 2003). We further extended Tasca and Tessone (2019)'s taxonomy to identify sustainability-related concepts referred to in the `ESG` criterion (see Figure 3).

3.2.2. Text analysis/language processing

We pruned the label hierarchy within the taxonomy for class balance, where specific labels such as PoW are replaced by broader categories such as Consensus to maintain focus on primary taxonomy components (Figure 3). To improve the NER model performance, which is sensitive to label consistency (Jeong & Kang, 2023; Zeng, Yu et al., 2021), we employed a

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Table 1. List of groups of entity types based on the extended taxonomy from Tasca and Tessone (2019) (see Figure 3)

Group entities	Description
Blockchain Name	The name of a blockchain system (e.g., Bitcoin, Ethereum, XRP Ledger), but also including other types of DLT, such as Hedera, IOTA, and Nano.
Consensus	Rules and mechanisms to ensure the immutability of transaction records (e.g., PoW, PoS, Blockchain, Hashgraph, Gossiping, Decentralized, Centralized, Distributed, Hierarchical, Latency, and Finality).
Transaction Capabilities	Information related to the details of transactions, such as Data Structure in the Blockheader, Transaction Model, Server Storage, Block Storage, and Limits to Scalability.
Native Currency Tokenization	Asset classes to pay for transactions within a blockchain system (e.g., BTC, ETH, XRP, HBAR).
Extensibility	Capabilities of Interoperability, Intraoperability, Governance (e.g., Decentralized Autonomous Organization (DAO), Alliance Mode, Open-source community mode), and Script Language of a blockchain ecosystem (e.g., Turing complete, non-Turing complete).
Security Privacy	Cryptographic methods to ensure data privacy, encryption, and security in a blockchain ecosystem (e.g., private key, public key, hashing, Sybil attack, 51% attack).
Codebase	Coding Language (e.g., Solidity, Vyper, Rust, JavaScript), Code License (e.g., GNU General Public License (GPL), MIT License, Apache License), and Software Architecture of a blockchain ecosystem (e.g., Monolithic, Polylithic).
Identity Management	Attributes to identify participants and their system access level.
Charging & Rewarding System	Cost and reward models for the operation and maintenance of blockchain systems (e.g., transaction fees, mining rewards, fee rewards, fee system, fee structure).
Identifiers	Information related to the token names (e.g., USDC, USDT), creators (e.g., Satoshi Nakamoto, Ripple, Leemon Baird, Circle, Coinbase), and purpose (e.g., stablecoins, utility token, security token).
ESG	Entities relevant to ESG issues regarding Blockchain (e.g., energy consumption, computing power).
Miscellaneous	Miscellaneous entities that are ambiguous in a given context and are relevant for the DLT topic but are not captured by any of the above categories.

systematic process for enhancing interlabeler consistency. This process involved correcting inconsistent labeling of entities. For example, "Sybil attack" was sometimes labeled as Consensus and, at other times, as Security Privacy. We resolved this inconsistency following each labeler's approval and using programmatic cleaning to ensure consistency for non-context-dependent labels. For "Sybil attack" specifically, we decided to label it as Security Privacy.

We applied text resampling for overlapping named entities that could fit into multiple categories, such as belonging to Blockchain Name and Native Currency Tokenization. This process involves duplicating text and assigning distinct entities to each copy, enhancing the capture of rare entities. This resampling strategy is beneficial, especially for data sets of modest size (Wang & Wang, 2022), improving model performance by accommodating diverse entity categories. Additionally, the duplication of training data enhances a language model's ability to learn from limited examples (Charton & Kempe, 2024; Muennighoff, Rush et al., 2023).

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3.2.3. Data set filtering

We applied a percentile-based filtering process to the corpus of publications based on the ESG and DLT classified named entities within the corpus. This method selects publications with substantial DLT and ESG content, using a threshold percentile to exclude marginally relevant papers. Seed papers were included to maintain foundational references. The filtering is represented as:

$$F = \{P_i : N(P_i) \geq \text{perc}_{10}(N(P))\} \cap \{P_i : D_{\text{DLT}}(P_i) \geq \text{perc}_{90}\} \cap \{P_i : D_{\text{ESG}}(P_i) \geq \text{perc}_{70}\} \cup S \quad (1)$$

where F is the final set of papers, P_i is an individual paper, $N(P_i)$ represents the total token count for paper i (with a minimum threshold at the 10th percentile to exclude potentially corrupted or incomplete documents), $D_{\text{DLT}}(P_i)$ and $D_{\text{ESG}}(P_i)$ are the DLT and ESG content densities respectively, and S is the set of seed papers. We define the percentile cut-offs for the DLT and ESG content densities by manually reviewing the ratio of NER labeled tokens to all tokens in the data set.

Our processing, filtering, and selection methodology involved (see Figure 2):

1. Processing: Excluding papers below the 10th percentile ($N(P_i) \geq \text{perc}_{10}(N(P))$) in the total token count as a quality control mechanism, eliminating documents that may have been improperly processed during PDF-to-text conversion or are too brief to contain meaningful analysis (e.g., below 100 tokens). We named these discarded papers as "parsing errors" in Figure 2.
2. DLT content filter ($D_{\text{DLT}}(P_i) \geq \text{perc}_{90}$): Computing DLT content density and retaining papers above the 90th percentile, ensuring a strong focus on DLT topics.
3. Filtering for at least the 70th percentile in ESG content density to confirm relevance to ESG ($D_{\text{ESG}}(P_i) \geq \text{perc}_{70}$).

Finally, we manually reviewed the selected filtered publications to validate the accuracy of their ESG/DLT content density and relevance.

3.3. Network Graphs and Entities Evolution

- *Keywords network*. We constructed a graph $G_{\text{keywords}}(V_{\text{keywords}}, E_{\text{keywords}})$ based on the full corpus of over 60,000 metadata records, with vertices V_{keywords} as individual keywords and edges E_{keywords} representing keywords co-occurrence within the same paper. We analyzed degree centrality with one-year rolling windows to observe the evolution and significance of specific keywords over time providing insights into how research from these keywords representing topics was foundational for the DLT field and facilitated the emergence of innovations.
- *ESG/DLT network*. We analyzed the ESG/DLT content density filtered citation network as $G(V, E)$, with papers as vertices V and citations as edges E. We performed temporal graph analysis using one-year time windows W_1, W_2, ..., W_n, following the rolling window approach from Hoadley, Bartolo et al. (2021), Steer, Arnold et al. (2024), and Steer, Cuadrado, and Clegg (2020). For each window W_i, we created a subgraph $G_i(V_i, E_i)$ and analyzed nodes, edges, average degree, and authority scores (using the HITS algorithm; Kleinberg, 2006) to determine temporal shifts and how publications rely on pivotal or existing work.

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Table 2. Labeled named entities for each category in the data set

Entity category	Number of entities
Blockchain Name	4,030
Consensus	14,530
Transaction Capabilities	5,120
Native Currency Tokenization	1,382
Extensibility	1,130
Security Privacy	5,892
Codebase	1,251
Identity Management	1,032
Charging & Rewarding System	794
Identifiers	1,058
ESG	2,068
Miscellaneous	1,140

- *Entity analysis*. We tracked the evolution of named entities in the ESG/DLT content density filtered citation network, consolidating variations of similar entities (e.g., unifying all forms of "Proof of Work" under "PoW") using lemmatization and programmatic grouping to accurately capture changes in entity prevalence.

4. EVALUATION

4.1. Taxonomy Labeling Result

Our NER data set organizes 39,427 named entities[4] from 80 manually labeled publications into a tree structure with 136 labels under 12 top-level categories (Figure 3a and Table 2). This structure facilitated a targeted analysis in our study. Table 3 provides examples from the data set.

4.2. NER Task Result

We fine-tuned four pretrained transformer models: the BERT base model (cased)[5], which uses a bidirectional encoder architecture (Devlin et al., 2019) (see Section 3.2 for more details); ALBERT base model v2[6], a lighter version of BERT with a cross-layer parameter sharing technique (Lan, Chen et al., 2019); DistilBERT base model (cased)[7], a distilled version of BERT

[4] An entity is counted from its beginning ("B-") to its end ("I-") using the IOB2 format (Tjong et al., 1999). See Table 3 for some examples.
[5] https://huggingface.co/bert-base-cased.
[6] https://huggingface.co/albert-base-v2.
[7] https://huggingface.co/distilbert-base-cased.

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Table 3. Training examples showing text spans with their labels ("Output") and corresponding IOB2 format (Tjong, Sang, & Veenstra, 1999), where "B-" marks the beginning of an entity, "I-" its continuation, and "O" denotes nonentity tokens, used for fine-tuning BERT-based models for our NER task

Text: *In this paper, the PoW consensus algorithm used in blockchains are analyzed in terms of difficulty, hash count, and probability of successful mining.*
Output: *In this paper, the ⟨Consensus⟩ consensus algorithm used in ⟨Identifiers⟩ is analyzed in terms of ⟨Consensus⟩, ⟨Transaction Capabilities⟩, and ⟨Transaction Capabilities⟩.*
IOB2: *O O O O ⟨B-Consensus⟩ O O O O ⟨B-Identifiers⟩ O O O O O ⟨B-Consensus⟩ O ⟨B-Transaction_Capabilities⟩ O ⟨B-Transaction_Capabilities⟩*
Text: *Given the fundamental challenges in uniting Bitcoin mining with renewable energy, along with the fact that energy use is not the only way in which Bitcoin impacts the environment*
Output: *Given the fundamental challenges in uniting ⟨Consensus⟩ with ⟨ESG⟩, along with the fact that ⟨ESG⟩ is not the only way in which ⟨Blockchain Name⟩ ⟨ESG⟩*
IOB2: *O O O O O O ⟨B-Consensus⟩ O ⟨B-ESG I-ESG⟩ O O O O O ⟨B-ESG⟩ O O O O O O O ⟨B-Blockchain_Name⟩ ⟨B-ESG⟩*
Text: *the Hedera Treasury will "proxy-stake" over two-thirds of the total number of hbars to nodes hosted by Council Members.*
Output: *the ⟨Extensibility⟩ will "⟨Consensus⟩" over ⟨Consensus⟩ of the total number of ⟨Native Currency Tokenization⟩ to nodes hosted by ⟨Extensibility⟩.*
IOB2: *O ⟨B-Extensibility I-Extensibility⟩ O ⟨B-Consensus I-Consensus⟩ O ⟨B-Consensus⟩ O O O O O ⟨B-Native_Currency_Tokenization⟩ O O O O ⟨B-Extensibility I-Extensibility⟩*
Text: *Using a variant of Shor's algorithm [162], a quantum computer can easily forge an elliptic curve signature that underpins the security of each transaction in blockchain and so breaking of ECC will affect blockchain in terms of broken keys, hence, digital signatures.*
Output: *Using a variant of ⟨Security Privacy⟩ [162], a ⟨Miscellaneous⟩ can easily forge an ⟨Security Privacy⟩ that underpins the security of each transaction in ⟨Consensus⟩ and so breaking of ⟨Security Privacy⟩ will affect ⟨Consensus⟩ in terms of ⟨Security Privacy⟩, hence, ⟨Security Privacy⟩.*
IOB2: *O O O O ⟨B-Security_Privacy I-Security_Privacy⟩ O O ⟨B-Miscellaneous I-Miscellaneous⟩ O O O O ⟨B-Security_Privacy I-Security_Privacy⟩ O O O O O O O O ⟨B-Consensus⟩ O O O O ⟨B-Security_Privacy⟩ O O ⟨B-Consensus⟩ O O O ⟨B-Security_Privacy⟩ O ⟨B-Security_Privacy⟩*

(Sanh, Debut et al., 2019); and SciBERT base model (cased)[8], a BERT-based model pretrained on a corpus of scientific publications (Beltagy, Lo, & Cohan, 2019). To ensure robust evaluation and prevent data leakage, we employed 5-fold cross-validation based on publication titles, ensuring that data from any single publication appeared exclusively in the training or test set for each fold. The training process consisted of 200 total epochs, distributed as 40 epochs per fold, with a learning rate of 5×10^{-5}, a training batch size of 32, and a validation batch size of 64. The maximum sequence length was 512 tokens.

The performance evaluation (Table 4) revealed that SciBERT (Beltagy et al., 2019) achieved the highest F1 score for our domain-specific NER task in DLT. This aligns with expectations, given SciBERT's pretraining on a large multidomain corpus of scientific literature, which improves its performance in downstream scientific NLP tasks (Beltagy et al., 2019) like ours.

[8] https://huggingface.co/allenai/scibert_scivocab_cased.

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Table 4. Detailed performance results using seqeval (Massias, Serret Avila, & Quisquater, 1999; Nakayama, 2018) after 5-fold cross-validation fine-tuning for BERT, Albert, DistilBERT, and SciBERT with the ESG/DLT NER data set, with relaxed and strict (exact) matched entities-level metrics

Model & fold		Relaxed entity-level metrics				Strict entity-level metrics			
Model	**Fold**	**Precision**	**Recall**	**F1**	**Accuracy**	**Precision**	**Recall**	**F1**	**Accuracy**
BERT	1	0.23	0.41	0.30	0.94	0.27	0.40	0.33	0.94
	2	0.63	0.63	0.63	0.97	0.74	0.62	0.67	0.97
	3	0.45	0.48	0.47	0.96	0.59	0.46	0.52	0.96
	4	0.54	0.50	0.52	0.96	0.60	0.49	0.54	0.96
	5	0.64	0.79	0.70	0.97	0.70	0.78	0.74	0.97
	Mean	0.50	0.56	0.52	0.96	0.58	0.55	0.56	0.96
Albert	1	0.30	0.31	0.30	0.94	0.34	0.31	0.32	0.95
	2	0.49	0.50	0.50	0.96	0.57	0.49	0.53	0.96
	3	0.40	0.43	0.41	0.96	0.50	0.42	0.46	0.96
	4	0.76	0.80	0.78	0.98	0.81	0.79	0.80	0.98
	5	0.80	0.83	0.81	0.98	0.85	0.83	0.84	0.98
	Mean	0.55	0.57	0.56	0.97	0.61	0.57	0.59	0.97
DistilBERT	1	0.26	0.38	0.31	0.93	0.31	0.37	0.34	0.93
	2	0.57	0.62	0.59	0.97	0.65	0.62	0.63	0.97
	3	0.52	0.66	0.58	0.96	0.65	0.65	0.65	0.96
	4	0.83	0.89	0.86	0.99	0.89	0.89	0.89	0.99
	5	0.62	0.80	0.70	0.97	0.72	0.80	0.76	0.97
	Mean	0.56	0.67	0.61	0.97	0.64	0.67	0.65	0.97
SciBERT	1	0.29	0.42	0.34	0.95	0.31	0.41	0.36	0.95
	2	0.63	0.66	0.65	0.98	0.71	0.66	0.68	0.98
	3	0.50	0.63	0.55	0.96	0.62	0.62	0.62	0.96
	4	0.76	0.86	0.80	0.98	0.84	0.86	0.85	0.98
	5	0.82	0.90	0.86	0.99	0.88	0.90	0.89	0.99
	Mean	0.60	0.69	0.64	0.97	0.67	0.69	0.68	0.97

Despite SciBERT's results, we ultimately chose DistilBERT, the second-best-performing model (Table 4), for processing our large data set of 24,539 publications. DistilBERT offers a reasonable balance of effectiveness and efficiency, operating 60% faster than BERT, and likewise SciBERT, during inference while retaining 97% of BERT's performance (Sanh et al., 2019). This speed and F1 score combination made DistilBERT the most suitable choice for our large-scale analysis of DLT literature.

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5. DISCUSSION

5.1. Transferability and Adaptability of the Methodology

5.1.1. Methodology framework transferability

Our methodology shows versatility in analyzing emerging technologies and their interdisciplinary implications. While we applied it to DLT[9] and ESG, the framework's structure makes it adaptable to various technological domains and their intersections, such as AI in healthcare or renewable energy applications in urban development, to name two.

5.1.1.1. Methodological framework. The methodology comprises five phases (Figure 1). Initially, we established a comprehensive corpus through data collection by selecting seed papers that represent seminal publications at domain intersections of interest via citation analysis and expert consultation, then systematically expanding this by traversal up to two levels of the seed paper's references (Section 3.1). The second phase focused on data labeling by first developing taxonomies that capture the domain intersections terminology, which serves as a guideline for labeling a NER data set (Sections 3.2.1 and 3.2.2). This labeled data set then supported the third phase, in which it is used to fine-tune BERT-based models for an NER task (Section 3.2), benchmarking and selecting the best-performing model (Section 4.2). The fourth phase applied the selected fine-tuned model to filter the corpus based on the named entities identified (Section 3.2.3). Finally, we conducted a manual literature review and temporal graph analysis to derive insights that track the evolution of concepts, technologies, and their interconnections.

5.1.1.2. Framework application example. To demonstrate the framework's transferability, consider its application to AI in healthcare research. The process would begin by identifying influential papers at the AI-healthcare intersection, followed by traversal of the references from these papers to build a comprehensive corpus. A specialized taxonomy would integrate medical terminology with AI concepts, enabling data set labeling for model fine-tuning. This fine-tuned model with the labeled data set would then facilitate the identification of specific medical AI technologies and their clinical applications. Finally, a manual literature review using temporal graph analysis, named entities, and citation network analysis would reveal the evolution of medical AI research and its healthcare impact.

5.1.2. Methodological advantages over other approaches

Our methodology (Section 3) of using NER for field-specific literature filtering is similar to recent LLM developments, such as Google DeepMind Gemini's demonstration of a systematic literature review. In Gemini's approach, terms like "Chip" and "CRISPR-Cas9" are searched in publication titles and abstracts to filter results[10]. This parallel highlights the significance of our work. However, Gemini faces limitations, such as potential hallucinations, that could undermine it for NER tasks and filtering of publications. Although supervised learning approaches outperform current LLM for NER tasks (Hu et al., 2024; Li et al., 2023), Gemini shows the potential of LLMs in few-shot learning for NER tasks.

In contrast to current LLM approaches, our methodology uses a domain-specific labeled NER data set and a fine-tuned pretrained language model to analyze full-text publications. This

[9] Section 6 explains DLT and some its key terms in more detail.

[10] See https://youtu.be/sPiOP_CB54A?feature=shared&t=64 for the prompt used in the demonstration.

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supervised learning approach provides a more structured and verifiable framework for literature analysis, making it particularly valuable for systematically examining emerging technical fields and their intersections with other domains. The methodology's emphasis on supervised learning and domain-specific training ensures greater accuracy and reliability (see Section 3.2 for more details) in literature review tasks while maintaining adaptability to various fields of study.

5.2. Findings

5.2.1. Keywords network

The keywords graph comprises 25,048 topics as nodes with 3,042,397 edges and 4,847,870 temporal edges connecting the topics across time. Figure 4 visually and quantitatively shows the evolution of the key areas like Cryptography, Peer-to-Peer, PoW, PoS, and smart contracts that laid the foundation for advancements in DLT at different points in time. Research in cryptography provided the foundations for using public and private key pairs for identity management (Chaum, 1981), elliptic curves (Miller, 1986), hashing, and Merkle trees (Merkle, 1980). Similarly, Peer-to-Peer research focused on network communication (Cohen, 2003; Dingledine, Mathewson, & Syverson, 2004), timestamping (Bayer, Haber, & Stornetta, 1993; Haber & Stornetta, 1991; Massias et al., 1999), and the need for solving the Byzantine Generals Problem (Lamport, Shostak, & Pease, 1982), vital for decentralized network functionality. The research related to Proof of Work delineated the first consensus system used in blockchain building in the works from the early 1990s of Dwork and Naor (1993) and the early 2000s of Back (2002) and Finney (2004).

The emergence of Bitcoin (Nakamoto, 2008) marked a significant convergence of these technologies. Its degree centrality as a research topic rapidly soared from 0.056 in the late 2010s to 1 in the 2020s, a growth of ~1,692.42%. This surge in interest, possibly fueled by the socioeconomic and political climate following the 2008 financial crisis (Mukunda, 2018; Schepisi, Di Napoli et al., 2021; Yaffe-Bellany, 2023; Zamani & Babatsikos, 2017; Zewde & Crystal, 2022) and Bitcoin's technological innovation, contrasted with digital money projects (Chaum, 1996; Dai, 1998; Szabo, 2005) in the 1990s and early 2000s that did not become as popular as Bitcoin in the same amount of time.

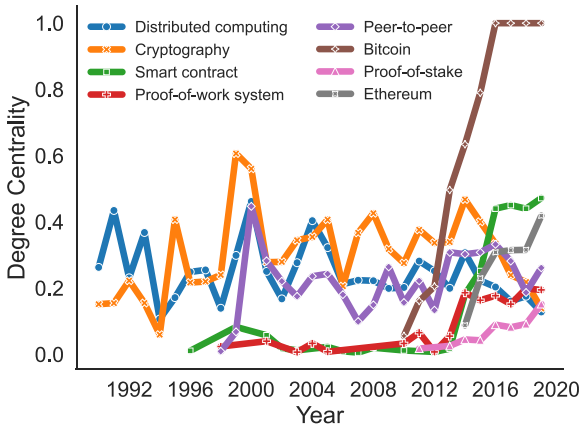


Figure 4. Evolution of research topics in the keywords graph from the full corpus of 63,096 publications.

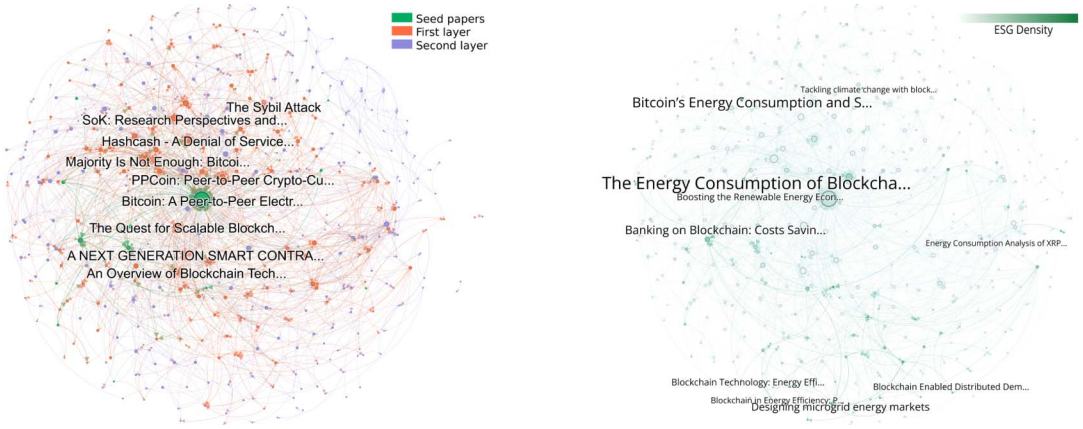
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Parallel to Bitcoin's rise, or inspired by it (Cornish, 2018), Ethereum emerged around 2014–2015, introducing two key technological innovations: the Solidity programming language and the Ethereum Virtual Machine (EVM) for smart contract development and execution, respectively (Buterin, 2014a). Ethereum's significance in the field grew substantially, with its degree centrality increasing from 0.088 in 2014 to 0.419 in 2020s (~376.64% growth). While smart contracts were conceptualized in the mid-1990s (Szabo, 1997), their adoption remained limited until Ethereum's implementation (Buterin, 2014a), which provided a comprehensive development and execution environment through Solidity and the EVM. This technological foundation catalyzed extraordinary growth in smart contract research interest, evidenced by a degree centrality increase from 0.011 in the mid-1990s to 0.472 in the 2020s (~4,299.04% growth). This research interest in smart contracts is reflected in efforts to extend its applicability, including legal aspects through Smart Legal Contracts (Roche, Hernandez et al., 2021) and their generation using NLP (Chen, Roche et al., 2023).

Amid these developments, Proof of Stake was introduced as an energy-efficient alternative to Proof of Work by King and Nadal (2012), which grew from 0.017-degree centrality since its introduction to 0.152 in the 2020s, (~768.68% growth), reflecting the growing interest in more sustainable consensus mechanisms.

5.2.2. ESG/DLT network

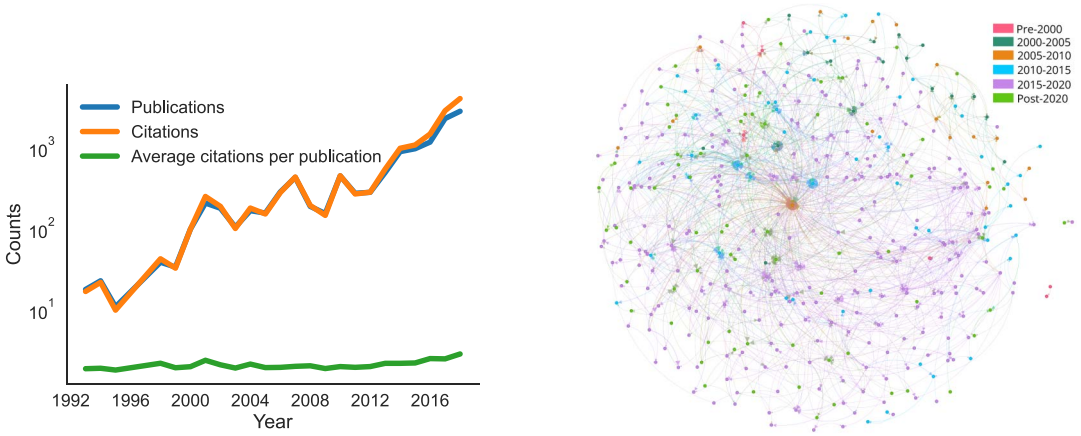
The ESG/DLT content density citation graph, derived from 505 papers, comprises 10,172 nodes (each node representing a citation), 15,898 edges, and 20,111 temporal edges connecting the publications over time. This graph facilitates narrowing down the publications specific to the ESG/DLT intersection (see Figure 2 and Section 3.2.3). The ESG/DLT network (Figure 5(a)) reveals a distinctive pattern in research evolution, particularly evident in the second layer of references, where multidisciplinary research demonstrates increased prevalence (Figure 5). This layer exhibits a notable concentration of ESG-focused investigations, especially regarding energy consumption and efficiency optimization of blockchain systems (Figure 5(b)). The peripheral references display an even broader interdisciplinary scope, which extends



(a) (b)

Figure 5. ESG/DLT network derived from 505 papers filtered by their content density of the ESG/DLT thematic (Section 3.2.3). (a) ESG/DLT network filtered by layers and authority scores. (b) ESG/DLT network filtered by the density of ESG-named entities. The network comprises 10,172 nodes (citations), 15,898 edges, and 20,111 temporal edges. Colors represent different layers (a) or content density for ESG entities (b), node sizes indicate authority scores, and only top nodes are displayed.

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(a) (b)

Figure 6. (a) Evolution of ESG/DLT network in terms of publications. (b) ESG/DLT network filtered by time windows and authority score distribution.

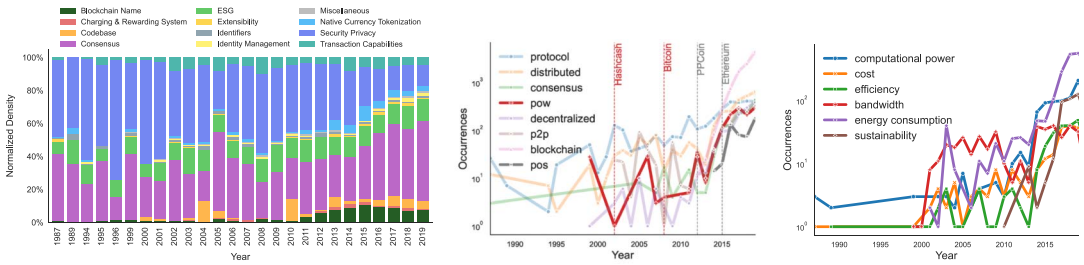
beyond traditional computer science boundaries into domains such as finance and environmental studies. This is likely because, as the DLT field develops, interdisciplinary research emerges.

The network (Figure 6) shows a publication surge between 2008 and 2011 of ~191.03%, aligning with Bitcoin's release and its subsequent influence on diverse DLT research areas, particularly in consensus mechanisms (Figure 7(a)[11]). Notably, Bitcoin's persistent and increasing significance in research attention (see also Section 5.2.1), despite the emergence of alternative solutions, demonstrates a "Lindy effect" (Goldman, 1964; Mandelbrot, 1983), where the protocol's longevity correlates with its continued academic interest. As Bitcoin gained more public attention after its appearance in 2008, often driven by price action, the research community evaluated this new technology, particularly Bitcoin's PoW (Biryukov & Pustogarov, 2015; Vukolić, 2016), for potential unaccounted environmental costs (de Vries, 2018; Mora, Rollins et al., 2018) or negative externalities (Jones, Goodkind, & Berrens, 2022; Papp, Almond, & Zhang, 2023).

Despite Bitcoin's prominent media coverage regarding energy consumption, our analysis of the ESG/DLT citation network (Figures 5 and 7(a)) reveals two significant patterns in research focus. First, ESG-related research has maintained a consistent proportional representation within the broader DLT literature rather than showing increased relative dominance (Figure 7(a)), with its primary focus remaining on energy consumption and efficiency optimization of blockchain systems (Figure 5(b)). Second, the research emphasis has evolved from an early focus on fundamental security and privacy considerations (Section 5.2.1) toward a growing focus on efficient (King & Nadal, 2012) and secure consensus algorithms (Douceur, 2002), and blockchain architectures (Bonneau, Miller et al., 2015; Vukolić, 2016). Curiously, we observe an increase in these studies investigating energy-efficient consensus mechanisms and scaling solutions during periods of rapid price appreciation, such as Bitcoin's bull runs (Canellis, 2023; Kucoin, 2024; Lang, 2024).

[11] The Blockchain Name entities identified before 2007, such as HashCash (Back, 2002), reflect the standard industry practice of using "Blockchain" as a generic term for any consensus-based DLT system, a convention we maintain in our taxonomic classification.

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(a) (b) (c)

Figure 7. Named entities evolution in the citation network. (a) Publications showing the normalized ratio of labels for each part of a branch of the taxonomy. (b) Top `Consensus'` named entities evolution. (c) Top `ESG`'s named entities evolution.

Post-2012, the network saw a marked increase in publications, especially after 2014, reflecting the impact of seminal works like PPCoin and Ethereum's whitepapers (Buterin, 2014a; King & Nadal, 2012). The growth in citations and publications from 2012 until the 2020s was ~901.75%. The increased interest in PoS as an energy-efficient alternative to PoW (Perez, Xu, & Livshits, 2020) is exemplified by Vitalik Buterin's early exploration of PoS before Ethereum's launch using PoW in 2015, which is evident in his 2014 Slasher algorithm post (Buterin, 2014b) and later posts that discuss PoS benefits (Buterin, 2014c), and the "nothing at stake" challenge (Buterin, 2014d).

Vitalik's early public discussions of PoS may have been one of the catalysts that encouraged further research into PoS (Figures 4 and 7(b)) before Ethereum's transition from PoW to PoS in 2022 (Ethereum Foundation, 2022). This is consistent with how Ethereum popularized smart contracts despite the research around smart contracts existing since the mid-1990s (Szabo, 1997), but remaining seemingly static as a topic of interest for the research community (Figure 4) until after Ethereum introduced the Solidity programming language and EVM (Section 5.2.1 and Figure 4).

DLT's thematic shift, coupled with the increasing prominence of terms like "decentralization," "blockchain," and "sustainability," underscores a multidisciplinary approach in the field (Figure 7(c)). The sustained interest in PoW, along with explorations into PoS and other consensus mechanisms, highlights the field's adaptability to environmental and scalability challenges (Figure 7(b)).

5.3. Limitations

Our literature review faces limitations, including potential biases in seed paper selection and a time lag in capturing recent publications, which may affect the comprehensiveness of our analysis. For instance, the choice of XRP's 2018 whitepaper (Chase & MacBrough, 2018) over the more cited 2014 edition (Schwartz, Youngs, & Britto, 2014) could underestimate its influence on the citation network. Similarly, recent works such as the 2018 Hedera whitepaper (Baird, Harmon, & Madsen, 2018) are omitted due to unavailable citation data.

Building the citation network predominantly from pre-2020 seed papers introduces a bias toward older publications, potentially overlooking newer research yet to achieve recognition (Figure 6(a)). While our methodology could theoretically filter citations to seed papers based on content density, our review focused solely on references within the seed papers, possibly limiting the thematic breadth.

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Regarding our NER data set performance, there is an expected noise level (Alfina, Savitri, & Fanany, 2017; Fetahu, Chen et al., 2023; Rucker & Akbik, 2023) that we tried to reduce by enforcing interlabeler consistency (see Section 3.2.2). Our data set, with 80 full-text manually labeled papers, is smaller than many general-purpose NER data sets (Loukas, Fergadiotis et al., 2022; Tedeschi, Maiorca et al., 2021; Tjong et al., 2003; Weischedel, Hovy et al., 2011) but is comparable to those in specialized domains like materials science (Cheung, Zhuang et al., 2024; Dagdelen, Dunn et al., 2024).

Our analysis contains 24,539 openly accessible publications from an initial corpus of 63,083 references (Figure 2). While this reliance on publicly available research introduces potential sampling limitations, several factors mitigate concerns about representativeness and comprehensiveness. First, empirical evidence demonstrates that open-access publications achieve broader academic engagement (Archambault, Amyot et al., 2014; McCabe & Snyder, 2014; Piwowar, Priem et al., 2018) and higher media attention (Schultz, 2021), suggesting they effectively capture significant research developments and may influence more research trends. Second, although only approximately 18% of journal articles are currently openly available (Piwowar et al., 2018), high-quality research authors increasingly seem to favor hybrid open-access journals (Gaulé & Maystre, 2011). Third, the DLT field's rapid evolution has elevated nontraditional literature, particularly whitepapers and industry publications, as crucial sources of technological developments. Seminal works such as the whitepapers for Bitcoin (Nakamoto, 2008), Ethereum (Buterin, 2014a), and PPCoin (King & Nadal, 2012) exemplify this trend. Our methodology addresses these considerations through two complementary approaches: (1) incorporating nontraditional academic literature to capture contemporary technological developments, and (2) leveraging metadata from paywalled publications in our broader analysis of the DLT field evolution (Section 5.2.1). This two-tier analytical approach (Section 3) ensures comprehensive coverage of DLT research developments while maintaining methodological rigor.

5.4. Future Work

Future work should focus on integrating metadata from different whitepaper versions, like XRP's 2014 and 2018 editions (Chase & MacBrough, 2018; Schwartz et al., 2014), and sourcing metadata from alternative databases for publications with missing information, such as Hedera's whitepaper (Baird et al., 2018). Additional research should also include regular updates to the taxonomy (refer to Table 1), expanding training data by annotating more seed papers and exploring various language model architectures.

Additionally, beyond the systematic literature review presented in this paper, our NER data set could also support other cross-domain literature review research, such as DLT applications across software development, cybersecurity, and governance frameworks. Also, it could enable relationship mapping between DLT concepts, supporting the construction of knowledge graphs. Finally, the data set could contribute to developing automated systematic literature review systems, particularly valuable given the rapid evolution of the DLT field. These potential applications emphasize our NER data set's versatility and value beyond this paper.

6. CONCLUSION

This paper addresses the scarcity of high-quality labeled NLP data for blockchain research by developing an NER data set for the ESG/DLT domains intersection from public sources. We conducted a systematic literature review analysis to demonstrate its utility and provide future

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research possibilities. Our analysis reveals the critical intersections of topics like Cryptography, Peer-to-Peer, Proof of Work, Proof of Stake, and smart contracts in the evolution of DLT. We observed Bitcoin's persistent dominance in research (a "Lindy effect"). Ethereum's significant impact on the adoption of smart contracts and PoS as an energy-efficient alternative to PoW. These insights, and the growing focus on energy-efficient consensus mechanisms, highlight the field's rapid adaptation to technological and environmental challenges.

We believe our work represents a step towards leveraging and facilitating the use of NLP to research rapidly evolving fields like DLT, enabling researchers, policymakers, and industry stakeholders to stay informed and make data-driven decisions.

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AUTHOR CONTRIBUTIONS

Walter Hernandez Cruz: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing—original draft, Writing—review & editing. Kamil Tylinski: Conceptualization, Data curation, Methodology, Visualization, Writing—original draft, Writing—review & editing. Alastair Moore: Conceptualization, Data curation, Methodology, Visualization, Writing—original draft, Writing—review & editing. Niall Roche: Data curation, Methodology, Software, Writing—review & editing. Nikhil Vadgama: Conceptualization, Data curation, Methodology, Resources, Writing—review & editing. Horst Treiblmaier: Conceptualization, Data curation, Methodology, Writing—review & editing. Jiangbo Shangguan: Data curation, Software. Paolo Tasca: Conceptualization, Resources, Supervision, Validation, Writing—review & editing. Jiahua Xu: Conceptualization, Resources, Supervision, Validation, Writing—review & editing.

COMPETING INTERESTS

The authors have no competing interests.

FUNDING INFORMATION

This research received no external funding.

DATA AVAILABILITY

The NER data set generated during this study is available at https://huggingface.co/datasets/ExponentialScience/ESG-DLT-NER.

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